SECTION A

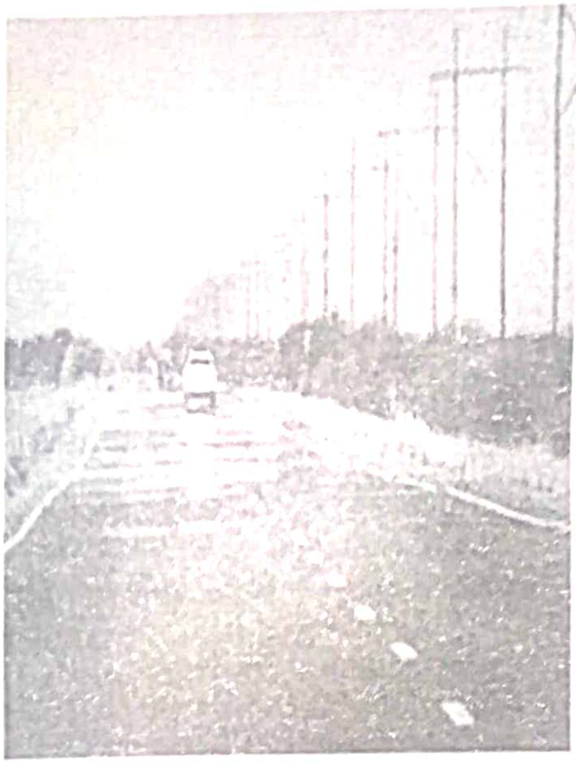
Answer **all** *the items in this section.*

Item 1

Ph.lip

One hot afternoon, a certain class walked along a tarmac road to check on a new swimming pool where they would have their P.E lessons. On the road, they saw a speeding car at a distance and its engine sounds kept on reducing as it moved further away. They also saw what looked like water near the car; but the water disappeared when they reached the spot where they had seen it and it reappeared to another spot ahead as they moved towards it. Later on, the weather changed and there was alight drizzle though the sun was still bright. In a certain direction, they observed a semi-circular distribution of colors in the sky.

On arrival at the swimming pool; the Pool- attendant warned them to be careful when they are to use the swimming pool because it appears shallow when filled with water.



The adventure ended in arguments because the learners had different views about the observations and experiences they had that day.

Task:

As a learner of physics, help the class to understand:

a. Why the sounds they heard from the car kept on reducing?
b. What looked like water near the car?
c. The process that leads to what they observed in the sky.
d. Why swimming pool appears the way the attendant told them?

ITEM 2.

Colleb

Recently, your female friend enrolled in an Astrophysics course in one of the outside countries. After a few days of settling in, she called her parents in Uganda by 9 am local time in that country. However, in Uganda, it was 8 pm. Her parents had just received the news that the full moon had been sighted using satellite communication, indicating the start of fasting for the holy month. During the phone conversation, she mentioned that her first lecture was about the life cycle of a star. This interested the parents who wondered how a star could have a life cycle and they were also curious about the significant time difference between the two countries.

Task:

As a physics learner, help the parents to understand;

a. What causes the differences in time between the two countries?
b. The different phases of the moon.
c. The life cycle of a star.
d. How it is possible to talk to her parents.

ITEM 3.

During a hospital tour, a liquid in a bottle spilled on one of the visitors. The visitor was told that the liquid that had spilled was a radio active material of activity 400 counts per second with a half-life of two days. He was told to self-isolate in the hospital until the count rate drops to 6.25 counts per second. The relatives of the isolated person tasked the hospital administration to come up with better ways to keep such materials.

Philip

Task:
Use your knowledge of physics to
a. Determine how much time the patient would take in the hospital
b. Explain to the relatives the dangers associated with radioactive materials.
c. Explain to the hospital staff how such materials should be handled.

SECTION B
Part I
Answer **one** *item from this part.*

Cullus

ITEM 4.

In a certain village, the roads are made of murram, with many potholes. On a certain rainy day, a driver of a car which has a mass of 1000kg travelling at $60kmh^{-1}$ sets off on a journey but along the way, the driver notices a big object on the road and suddenly brings the car to rest. In the process, the car skidded off the road and got stuck.

Whenever the braking force is applied in such a manner, about 60% of the kinetic energy is expended in the brake drum. The material of the drum and brake pads can absorb heat up to 150kJ of heat energy, beyond which the tyres will burn.

Task:

As a learner of physics, help the driver;

a. Understand how to avoid similar occurrence.
b. To understand why the car tyres did not burn.

ITEM 5.

A tourist who weighs 70kg, before climbing a mountain is availed with two sets of shoes; one with sharp spikes and the other with flat soles.

While climbing the mountain, the tourist follows a slippery road and had difficulty in climbing. Along the way, the tourist records a barometric reading at location A and B as 75cmHg and 71.7cmHg and wonders how far apart they are. The tourist felt very exhausted when arriving at location B and at the same time, he had nose bleeding. This scared him more.

Later he also had difficulty in having the food ready.

Hint;

Density of air= $1.25kgm^{-3}$

Density of mercury = 1.36×10^{4} kgm^{-3}

Task:

As a learner of physics, help the tourist understand why;

a. There was difficulty in climbing the mountain and how far apart the two points were.
b. He was very tired and the cause of nose bleeding.
c. There was difficulty in getting the food ready.

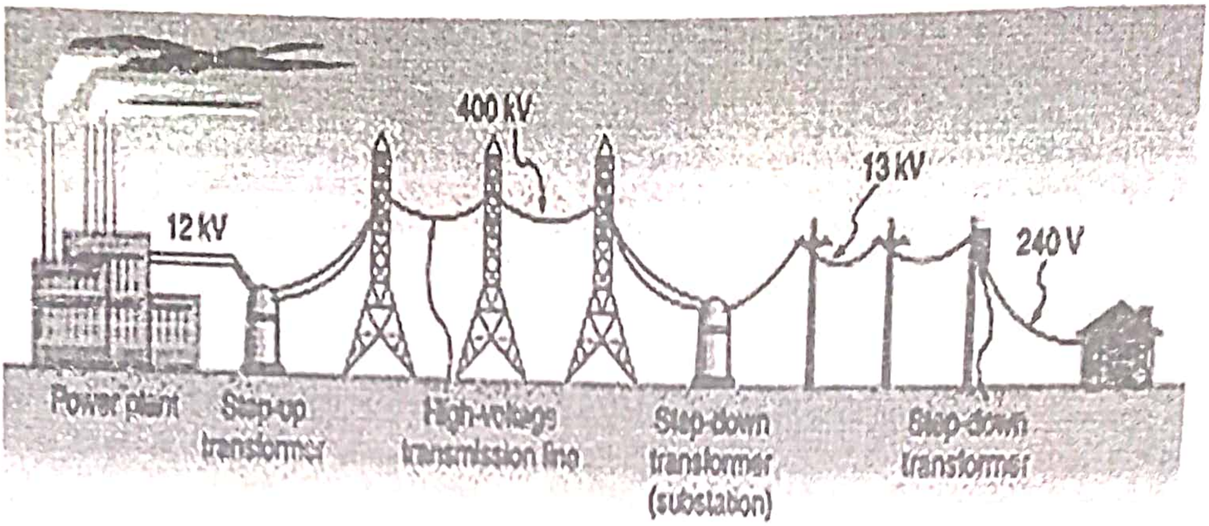
Part II

Answer **one** *item from this part.*

ITEM 6.

e

In a certain village, electricity at a substation is transmitted at 13 kV with a current of 0.05 A using thick aluminium wires for use inside a house at 240 V.



One of the villagers wanted to install two television sets (TVs) in his house. He approached a technician who told him that the TVs operate on direct current of 4A and 20% of the electrical energy is lost during voltage change. The villager did not understand what he meant and wondered if the TVs will still work.

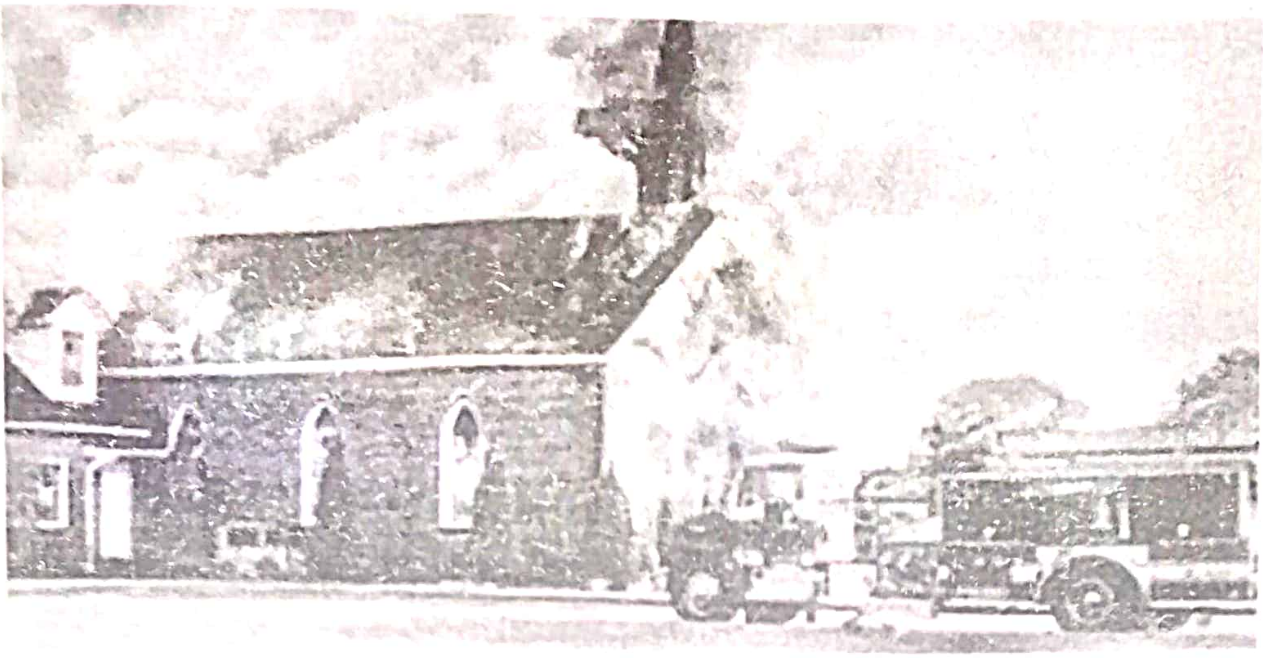
Task

Use your knowledge of Physics to help the villager;

a. Understand how the voltage is changed from 13 kV to 240 V.
b. Why thick aluminium wires are used during power transmission?
c. Determine if television sets will still work.

ITEM 7.

Residents in a community witness a church set on fire soon after hearing thunderstorm during a heavy rain. They were shocked by the amount of fire that erupted and destroyed a lot of property and wondered whether the cause was due to witchcraft.



In order to mobilize the community to contribute materials that may be used for the reconstruction of the church, the chairperson LC 1 used a portable loud speaker rated 36W, 12V. However, a resistor in the loudspeaker got burnt and needed replacement so that it would work properly.

Task:

As a learner of physics;

a. Help the residents to understand that the fire that erupted and destroyed the church was not due to witchcraft.

b. Advise the community to reconstruct the new church that would not be burnt again during thunderstorms.

c. Help the residents to identify the value of the appropriate resistor that was burnt so that the loud speaker can work properly.

END

535/1
PHYSICS
Paper 1
July/August 2025
2 ½ hours

ASSHU ANKOLE JOINT MOCK EXAMINATIONS 2025

Uganda Certificate of Education

PHYSICS

Paper 1

(Theory)

2 hours and 30 minutes

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of* two *sections;* A *and* B. *It has a total of* seven *examination items.*
- *Section* A *has* three compulsory *items.*
- *Section B has two parts; I and II. Respond to* one *item from* each *part.*
- *Respond to* five *items in all.*
- *All responses must be written in the booklets provided.*
- *Any additional item will* not *be scored.*

Physics 535/1

ASSHU ANKOLE JOINT MOCK EXAMINATIONS 2025

Uganda Certificate of Education

SCORING GUIDE

SECTION A

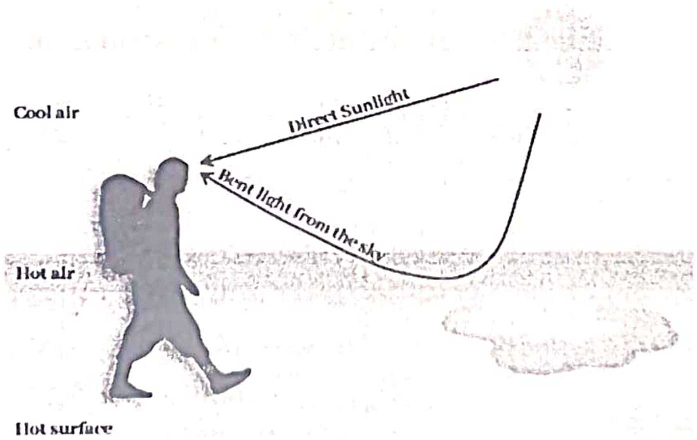
ITEM 1

a. The sounds from the car kept reducing because;

04 When the sounds from the car move away from an observer, the sound waves stretch out, increasing their wavelength.✓This results in a decrease infrequency✓and a reduction in loudness✓ as the sound energy spreads over a larger area. Thus, the farther the car moves, the quieter the sound becomes.✓

- The intensity of sound varies with distance ✓
- The intensity decreases with increasing distance ✓
- As the car moves from the learners, the distance between it and learners increases. This means that the sound from the car keeps on reducing in intensity as distance increases ✓
- This is why the sound heard from the car kept on reducing

b. What looked like water near the car was a mirage,✓ which is caused by refraction of light.✓

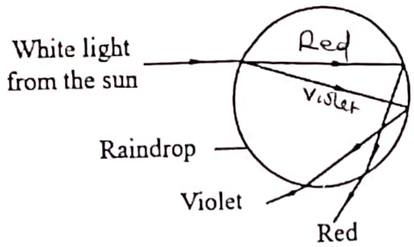


✓

On a hot tarmac road, the air near the ground is much hotter than the air above.✓Hot air is less dense, causing light from the sky to bend ✓(refract) upwards as it passes through these layers. This creates an illusion of a reflective surface (like water) on the road,✓ which disappears when approached because the viewing angle changes. 06

c. ~~The semi-circular distribution of colours in the sky~~

c) This was a rainbow, ✓formed due to dispersion, refraction, and reflection of sunlight in water droplets.



White light from the sun undergoes dispersion✓ as it enters into the raindrops of water in the sky. Total internal reflection✓ takes place at the opposite side of the raindrop and different colours ✓emerge from the raindrop after refraction✓

colours of the spectrum

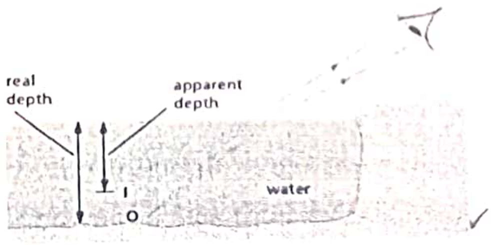
07

The semi-circular shape occurs because

- only light entering droplets at a 42° angle from the observer's shadow point reaches their eyes.✓
- the observer is at the surface of the earth., · which is spherical
  Rain droplet is spherical

  (Any one correct reason)

d. The swimming pool appears shallower than it really is due to refraction✓of light at the water-air boundary.



Rays of light from a point O at the bottom of the pool are refracted away from the normal✓ at the water surface because they are passing into an optically less dense medium, i.e. air. On entering the observers' eye, they appear✓to come from a point I that is *above* Q; I is the virtual image of O formed by refraction at the apparent depth of the pool that is less than its real depth.

05

(22 scores)

ITEM 2

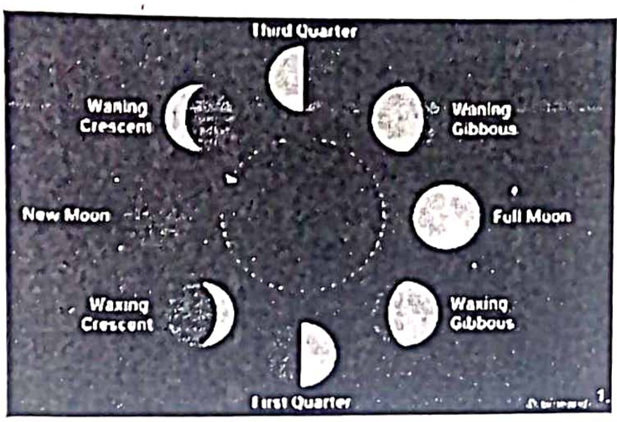
a. The time difference is due to earth's rotation.✓The earth's rotation makes different places experience day and night at different times, leading to time zones.✓Earth rotates 360° in 24 hours, meaning it moves 15° per hour. The world is divided into 24 time zones,✓ each roughly 15° of longitude apart.

time zone can also be scored on rotation of 360 in 24hrs

04

b. The Moon's phases are caused by its position relative to the earth and sun.✓

- The phases of the moon are caused by the rotation of the moon around the earth
- The moon is a natural satellite around the earth



✓

Note:
- ~~score~~ Put all the points in the diagram with correct labelling against the correct shape
- score the list alone and leave one score for the diagram.

New Moon – Moon is between Earth and Sun (not visible).

Waxing Crescent – A sliver of the Moon becomes visible.

First Quarter – Half of the Moon is lit (right side).

Waxing Gibbous – More than half is lit.

Full Moon – Earth is between Sun and Moon (fully lit).

Waning Gibbous – Light starts decreasing.

Third Quarter – Half lit (left side).

Waning Crescent – A shrinking sliver before the next New Moon.

06

5- 8 phases award 4 scores

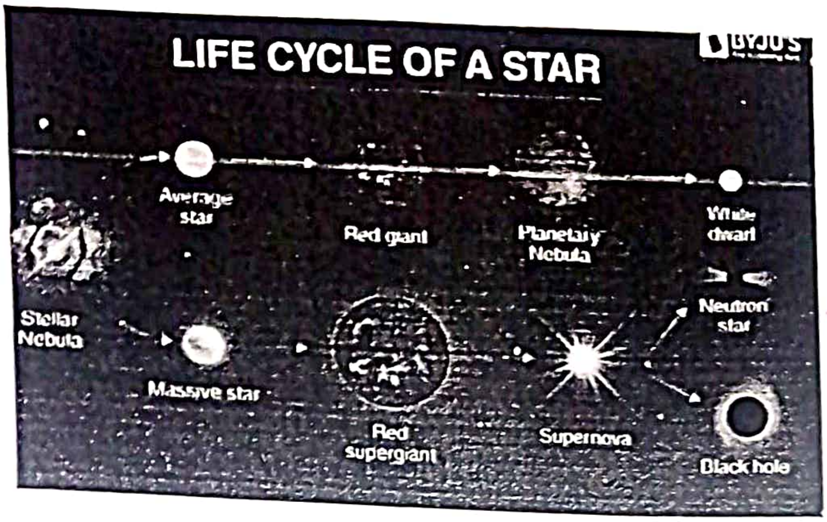
3-4 phases award 2 scores

1-2 phases award 1 score

04

c. The life cycle of a star

Stars go through stages based on their mass:



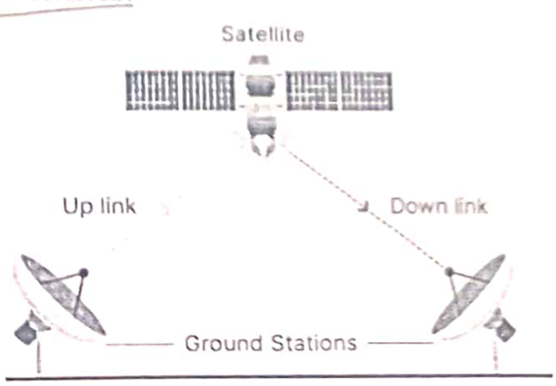
A giant gas cloud (nebula) collapses under gravity.✓It heats up and forms a proto star.✓ When the core gets hot enough nuclear fusion begins turning hydrogen into helium and releasing energy. The star spends most of its life in this stage. Our sun is in this stage

Small stars (like the Sun) expand into red giants.✓Massive stars become super giants.✓Small stars (like the Sun) shed outer layers to become planetary nebula and the core becomes a white dwarf which later cools into a black dwarf.✓ 07

Massive Stars:

Explode as a supernova and the core collapses into a neutron star or black hole.✓

d. She was able to talk to her parents by the help of communication satellites.✓



When a user on a satellite phone initiates a call, the signal is transmitted✓ to the nearest satellite✓ orbiting the earth. The satellite receives the signal, amplifies it,✓ and then retransmits it to a ground station✓ on earth. The ground station connects the satellite signal to the regular telephone network, allowing the call to be routed✓ to the intended recipient. The call continues to be relayed through the satellite until the conversation is over. 07

(24 scores)

$\frac{N_0}{N} = 2^{t/t_{1/2}}$ | $2^6 = 2^{t/2}$ ✓ $= 12$ days

$\frac{400}{6.25} = 2^{t/2}$ | $t = 6 \times 2$

$64 = 2^{t/2}$ | Concn ✓ Accept ✗

– Formula

– table

ITEM 3

a. 400 $\xrightarrow{2days}$ 200 $\xrightarrow{2days}$✓ 100 $\xrightarrow{2days}$ 50 $\xrightarrow{2days}$✓25 $\xrightarrow{2days}$ 12.5 $\xrightarrow{2days}$ 6.25✓

$$2 \times 6 ✓ = 12 \text{ days.} ✓$$

The patient would take 12 days in the hospital.✓

b. Dangers associated with radioactive materials
   - Cause radiation burns✓
   - Tissue damage✓ fishing 03
   - Cause cancer and tumors✓
   - Genetic damage to the genetic cells

   (Score any correct 3 risks)

c.. How such materials should be handled
   - Radioactive sources should be handled with forceps and kept in led boxes.✓
   - When using radioactive liquids, plastic or stainless trays should be used to contain any spill and should be kept in unbreakable containers.✓ 03
   - Gloves must be worn when unsealed source is being used to avoid direct contact with the substance.✓ fishing
   - Radiation badges containing photographic film should be worn to monitor exposure to radiation.

   (Score any correct 3 ways)

   (12 scores)

SECTION B
PART 1

ITEM 4

a. The car skidded off the road because the road was slippery.✓ The slippery road reduced the friction,✓thus the grip between the tyres and the road was less. 13 pts

Bringing the car suddenly to a stop is not possible because of inertia.

$$momentum = m \times v$$ ✓

$= 1000 \times \frac{60\times1000}{3600}$ ✓✓ one score for substitution, one score for conversion of speed.

2nd score for conv

$$= 16666.7\ kgms^{-1}$$ ✓✓

The car continued in its state of motion on the slippery road, and because of momentum,✓ it eventually came to a halt. This was due to air resistance.

The momentum of 16666.7kgms^{-1} is so big, hence making it hard to stop the car instantly. ✓

How to avoid such occurrences

- Driver needs to drive at a low speed so that the tryes are continuously in contact with the road. ✓Thus, making the car exert a larger weight on the road which increases friction.
- The driver should check that the tryes are not worn out, or use tyres with treads ✓otherwise friction will be low and will have higher chances of skidding off. Any 4
- adding more stones on the road ✓
- Tarmacking the road ✓

b. $total\ kinetic\ energy = \frac{1}{2}mv^2$ ✓

$$= \frac{1}{2} \times 1000 \times 16.6667^2$$ ✓

$$= 138889.4J$$ ✓

$$60\%\ of\ kinetic\ energy = \frac{60}{100} \times 138889.4$$

05

$$= 83333.64J$$ ✓

$Since\ 83333.64J < 150000J, the\ tyres\ will\ not\ burn,$ ✓ $hence\ the\ vehicle\ is\ safe.$

(18 scores)

ITEM 5

a. The tourist must have worn flat✓ shoes which have a large area of contact✓. This exerts less pressure✓ which decreases on the grip on the ground resulting to sliding✓. Hence making it difficult to climb the mountain.

How far apart A and B were.

$P_A - P_B$ = pressure due to air✓

$\frac{75}{100}$ ✓ $\times 13600 \times 10 - \frac{71.7}{100} \times 13600 \times 10 = h \times 1.25 \times 10$ ✓

$$102000 - 97512 = 12.5h$$
$$12.5h = 4488$$

10

$h = 359.04$✓ m✓

The two points were 359.04 m apart.✓

b. As he moves up the mountain, he does work against gravity.

$$work\ done\ by\ the\ tourist = mgh$$

$= 70 \times 10 \times 359.04$✓

$= 251{,}328\ J$✓

He does a lot of work of 251,328 J, resulting into loosing energy hence getting tired.✓

The cause of nose bleeding

06

The tourist experiences a reduction in the external pressure from 102000 Pa at A to 97512 Pa at B.✓ But his blood pressure remains the same. The excess internal pressure✓ makes his blood capillaries to burst✓ hence nose bleeding.

c. The cooking will take longer and require a lot of heat. As the person moves up the mountain, atmospheric pressure reduces.✓ This implies that water will boil at a reduced temperature✓ (lower boiling point) hence suppling less heat than that needed to cook the food. This makes it difficult for the food to get ready.

02

(18scores)

PART II

ITEM 6

OR
Diagram
Explanation
-ident freq
other explanation

a. The change in voltage from 13 kV to 240 V is achieved using a step-down, transformer ✓Step-down transformers have more turns in the primary coil than the secondary✓, reducing the voltage to a safer level for household use. Alternating Current in the primary coil creates a changing magnetic field. This induces a voltage in the secondary coil.

04

b. Thick aluminium cables are used in power transmission for the following reasons:

- Low Resistance✓ – Thicker cables have a larger cross-sectional area, reducing resistance which minimizes power loss ($P\ loss = I^2R$). fusing
- Aluminium is cheaper and lighter, ✓ –making it cost-effective for long-distance transmission.
- Heat dissipation – Thick cables can handle high currents without overheating, ensuring efficiency and safety.
- Reduced voltage drop – Lower resistance means less voltage is lost over long distances.

07

(Score any ~~correct~~ cor 1 2)

(c)

$$Power\ transmitted = IV ✓$$

$$P = 0.05 \times 13000 ✓$$

$$= 650 ✓ W ✓$$

b 10

After Step-down transformer **(240 V)**

$$20\%\ enregy\ is\ lost, so\ usable\ power\ becomes$$

$$100\% - 20\% = 80\%$$

score 80%
or when implied
i.e $\frac{80}{100} \times 650$.

$$P = \frac{80}{100} \times 650 ✓$$

$$= 520W ✓$$

Again from $P = IV$

$$520 = I \times 240 ✓$$

$$I = 2.167\ A ✓$$

Since the TVs *operate on direct current of 4A, yet the transformer supplies current of 2.167A on AC, the* TVs *will not work.* ✓

(14 scores)

ITEM 7

a. The fire was not caused by witch craft, it was lightning.✓Lightning is a giant spark of electricity in the atmosphere between charged clouds.✓During a thunderstorm, charged clouds induced an opposite charge on the church✓and

05

repeled like charges through the church. ✓As a result, an electric wind was created causing sparks✓ igniting fires on the church.

b. To prevent the new church from being burnt again;

- Install a lightning conductor.✓A lightning conductor is mounted at the highest point of the church, connected to a ground wire. This helps to direct ~~lightning~~ excess safely into the earth, ~~by~~ by passing the building.
- Use fire-resistant materials such as metal roofing instead of thatch/wood, concrete/brick walls.✓
- Ensure all wiring is properly grounded to avoid sparks during storms.

Pick any (two)

02

c. The loudspeaker is rated 36W at 12V. To find the resistor's value:

$$P = IV ✓$$

$$36 = 12\,I ✓$$

$$I = 3 ✓A ✓$$

$$From\ V = IR$$

$$12 = 3R ✓$$

07

$R = 4\ \Omega$✓

The resistor should be 4Ω.✓

(14 scores)

END